VAIBHAV BAJPAI

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Work Experience

BuzzyBrains Software (I) Pvt. Ltd. | Pune | Senior Software Engineer ***May 2025 – Present***

- Led full-stack development of healthcare platform with React Native mobile app and React web application, implementing comprehensive RBAC across 5 roles (Doctor/Patient/Receptionist/Clinic/Super Admin); architected role-specific UI flows, permission-based feature access, and JWT-based authentication with OTP verification.
- Designed RESTful API architecture with Node.js/Express and PostgreSQL featuring granular RBAC logic, role-based endpoint authorization across 5 user types, JWT middleware, pagination, request validation, and centralized error handling; achieved 100% Postman test coverage.
- Built dynamic QR code generation system with backend hash-to-ID mapping and react-native-qrcode-svg rendering on mobile; integrated thermal printer compatibility for receptionist workflows enabling sub-5-second patient check-ins.
- Engineered multi-channel notification pipeline (Email/SMS/WhatsApp) with role-triggered templates for OTP delivery, appointment confirmations, and automated reminders; reduced patient no-shows through real-time, role-aware communications across mobile and web platforms.
- Developed Super Admin web portal (React) and mobile receptionist interfaces with hierarchical CRUD operations for Doctor/Patient/Receptionist/Clinic management, patient/dependent relationships, bulk data cleanup, appointment editing, and dynamic role provisioning across 50+ clinic profiles.
- Owned end-to-end CI/CD pipeline and team coordination: automated React Native APK/IPA builds, React web deployments, backend releases, database migrations with role-based seed data, environment provisioning, and production incident resolution with <2-hour SLA.

OneLab Ventures | Pune | SDE-1 ***July 2024 – May 2025***

- Developed and maintained full-stack applications for both web and mobile using the MERN stack and React Native.
- Integrated Google AdMob for revenue generation in a globally launched mobile app, optimizing ad placements and ensuring a seamless user experience.
- Designed and implemented critical APIs for app functionality, addressing high-volume requests and ensuring secure data flow.
- Collaborated with international clients, translating their requirements into efficient solutions and delivering high-quality features.
- Worked extensively with push notifications and In-app notifications, improving user engagement and retention.
- Developed and deployed a fully functional website using AWS services like EC2, S3, and CloudFront to ensure scalability, reliability, and optimal performance.

Perla IT Soln. International Pvt. Ltd, Nagpur | Software Engineer ***July 2023- July 2024***

- Developing WYSAX School Management Project: Utilizing ReactJS, NestJS, and PostgreSQL on AWS to streamline administrative tasks and enhance operational efficiency. Designed user-friendly interfaces with React and Material UI, supporting multiple languages (English, French, Arabic) using i18n for localization.
- Implemented role-based access control for secure data management and user-specific task permissions.
- Engineered multi-organization support, enabling seamless switching between entities for school administrators.
- Developed exam creation, student mark recording, and report card generation features for accurate performance tracking.
- Utilized PostgreSQL for efficient data storage and TypeORM for frontend-backend data communication via RESTful APIs built with Nest.js decorators and middleware.

Perla IT Soln. International Pvt. Ltd, Nagpur | Software Engineer Intern ***Jan 2023 – June 2023***

- Developed a school management system for international markets as a Software Engineer intern.
- Modernized legacy codebases and created an exam-taking platform with user authentication and result tracking.
- Collaborated throughout the software development lifecycle to meet project goals and ensure user satisfaction through extensive testing.

Projects

ProShop – Electronic Shopping Web Application: Developed ProShop, an electronic shopping website, using React, Node.js, Express, and MongoDB. Integrated secure authentication, optimized database performance, and ensured smooth communication between the frontend and backend. Designed for seamless browsing and purchasing, allowing users to conveniently shop for products online. Rigorously tested for reliability and designed with responsive principles for cross-device compatibility.

Daily Expense Tracker System: Developed a robust expense tracker with HTML, CSS, JavaScript, and PHP. Implemented CRUD operations, AJAX for data manipulation, and Chart.js for dynamic visualization. Enabled users to track their expenses efficiently and view them in an organized manner for better insights. Ensured data integrity and security, generated detailed reports with PHP, and deployed on a

LAMP stack for full-stack proficiency. Employed responsive design for cross-device performance.

Skills

- **Programming Languages** - C++, C.
- **Web Development** - HTML, CSS, Bootstrap, JavaScript, React JS, React Native, Expo, Node JS, MongoDB, SQL, Nest JS.
- **IT Constructs -** OOPs, Operating Systems, DBMS, Computer Networks.
- **Development Tools-** Microsoft Visual Studios Code, Git and Github.

Education

- **Bharati Vidhyapeeth College of Engineering, Pune, Maharashtra, India** ***2019-2023***
  B.Tech | CSE | 9.2 CGPA (A+ Grade)
- **Kendriya Vidyalaya no.2, AFS, Pune, Maharashtra, India |** *CBSE (Class XI)* ***2018-2019***
- **Kendriya Vidyalaya no.1, Raipur, Chhattisgarh, India |** *CBSE (Class X)* ***2015-2016***

Positions of Responsibility

ACSI - Alliance for Computer Science Integration - *Secretary* ***June'22 – June'23***
Rotary Internationals | *Professional Development Director* ***Mar'21 – Mar'22***